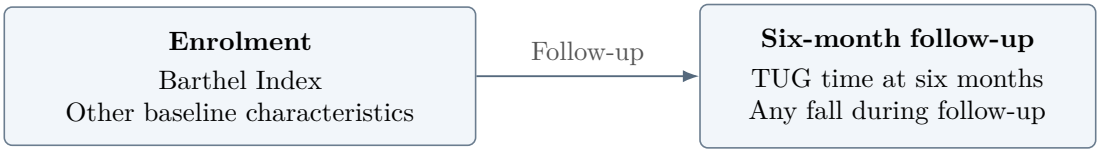
Functional Independence, Mobility and Falls

Regression | Splines | Missing data

Clinical question

In a fictional observational cohort study, 600 adults aged 65 years or older living in their own homes without formal home-care assistance were followed for six months to assess mobility and falls. The researchers want to investigate whether baseline functional independence is associated with subsequent mobility and the probability of falling, and to describe the shape of these associations.

Functional independence is measured at enrolment using the Barthel Index, which assesses independence in activities of daily living. Higher scores indicate greater independence. Mobility is measured at follow-up using the Timed Up and Go (TUG) test: the time taken to stand up, walk three metres, return, and sit down. Longer times indicate poorer mobility. The two measures capture related but distinct aspects of functioning. Age, previous falls, and other baseline characteristics are relevant when interpreting the Barthel associations.



Each participant contributes one record, identified by `participant_id`. Assume complete follow-up without death or loss to follow-up. The simulated dataset `HOME_FALLS.csv` contains the following data:

Outcomes	
`followup_tug`	TUG test after six months, in seconds. Longer times indicate poorer mobility.
`fall_followup`	At least one fall during the six-month follow-up (1 = yes, 0 = no).
Main explanatory variable	
`baseline_barthel`	Barthel Index at enrolment (0–100, recorded in five-point increments). Higher scores indicate greater independence in activities of daily living. Some scores are missing.
Other baseline variables	
`age`	Age at enrolment, in years.
`sex`	Recorded sex (Female or Male).
`living_alone`	Lives alone at enrolment (1 = yes, 0 = no).
`previous_fall`	At least one fall in the six months before enrolment (1 = yes, 0 = no).
`walking_aid`	Uses a walking aid at enrolment (1 = yes, 0 = no).

The two main research questions are:

- **How is baseline functional independence, measured by the Barthel Index, associated with mobility after six months?**
- **How is baseline functional independence associated with experiencing at least one fall during follow-up?**

Your task

Please analyse the dataset and report on the association of baseline functional independence, measured by the Barthel Index, with mobility at six months and with experiencing at least one fall during follow-up. The principal aim is to describe the shape, magnitude, and uncertainty of these associations, both before and after adjustment for other recorded baseline characteristics.

Explore and summarise the data, including the distributions of the variables and the extent of missingness. Include a baseline table describing participants overall and separately for those with and without previous falls.

Perform both unadjusted and adjusted regression analyses for each outcome. Make a reasoned choice of regression model and adjustment variables. Pay particular attention to the shape of the Barthel association, considering both linear and spline representations. Use suitable methods covered in the course to assess the evidence of association and to compare model specifications. State the assumptions of your chosen models and assess whether they are reasonable for these data. Use the same complete-case observations for relevant model comparisons.

Select a final model for each outcome and justify your choices. Discuss the likely mechanism underlying the missing Barthel assessments and use multiple imputation to account for them. Explain and justify your imputation approach, refit the two selected models without changing their specifications, and pool estimates and uncertainty using Rubin's rules. Compare the findings with those from the complete-case analysis.

Present the Barthel associations using fitted curves with 95% confidence intervals and meaningful score comparisons, rather than interpreting individual spline coefficients. For a realistic baseline profile, compare predicted outcomes at Barthel scores of 70 and 90, or justify alternative scores if these combinations are poorly supported. Report predicted mean TUG times and fall probabilities with 95% confidence intervals, together with the predicted differences. Confidence intervals for the differences are not required.

Explain what the findings imply for the two research questions. Distinguish associations from causal effects and remember that mobility measured only at follow-up does not establish improvement or deterioration. The analytical decisions and the methods used to support them are yours to justify; extensive searches through alternative models are not required.

Report and submission

Work in pairs. Report your findings in the style of a Results section of a scientific paper, preceded by a brief methods paragraph and followed by a short concluding interpretation. Present a coherent account, not separate answers to every instruction. Assessment focuses on the justification of your decisions, the main findings, and their interpretation, rather than the amount of output or whether results are statistically significant.

Submit a textual report as a PDF, not slides. The report must not exceed **1,200 words and four A4 pages**, with at most **three tables and figures combined**. A multipanel figure counts as one figure. Both length limits apply. The word count includes captions but excludes numerical table entries. The page limit includes everything. Do not submit an appendix or additional narrative material.

Submit a reproducible R or Python script separately, containing the analyses and checks not selected for the report. The baseline table and selected main findings belong in the report. The report is the primary assessed submission and the script may be checked to verify results.

Upload your results into the canvas-site before Monday 9 November 2026, 09.00. Don't forget to write your names in/on the presentation.

Good luck!